

Run 07 | TEA

Global (including India)

IV + controls

Target Total early-stage entrepreneurial activity; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 154,420 (11.50%) **Case time** 49m 26s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9937). Highest F1: Random Forest (0.8873). For XGBoost, adding controls changed ROC-AUC by +0.0023 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9937	0.9527	97.36	88.53	88.49	0.8851	18s
Random Forest	0.9937	0.9520	97.35	86.73	90.84	0.8873	2m 46s
CatBoost	0.9934	0.9507	97.30	88.16	88.39	0.8827	2m 47s
H2O GBM	0.9923	0.9402	97.02	85.61	89.03	0.8728	44s
Lasso	0.9916	0.9350	96.93	85.15	88.77	0.8692	13m 40s
Elastic Net	0.9916	0.9350	96.93	85.13	88.78	0.8692	11m 33s
Logistic Regression	0.9914	0.9342	96.91	85.15	88.61	0.8685	2m 16s
Ridge	0.9910	0.9307	96.84	84.87	88.30	0.8655	10m 10s
Decision Tree	0.9893	0.8947	96.82	85.63	86.90	0.8626	12s
Naive Bayes	0.9016	0.4368	87.39	46.91	73.41	0.5725	4m 05s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	69,979.51	45.62
2	Established business owner	28,441.92	18.54
3	Occupational status	11,814.93	7.70
4	Age	3,730.48	2.43
5	Self-efficacy	3,610.25	2.35
6	Knowing entrepreneurs	3,120.79	2.03
7	Education	1,679.65	1.09
8	Safety and security	1,603.80	1.05

Share is the percentage of total forest impurity importance across all 66 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	1,679.65	1.09
2	Safety and security	1,603.80	1.05
3	Living conditions	1,555.64	1.01

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 07_obj2_global_iv_cv_TEA. Scores and importance are taken directly from saved CSV outputs.